Machine Learning Notes

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ML (E-P-T)

- **E** → Experience
- **P** → Performance
- **T** → Task

Types of Machine Learning

1. Supervised Learning (Labelled Data)

Input Set + Output → Learning Algorithm → Model

New Input (X) → Model → Output (Y)

2. Unsupervised Learning (Unlabelled Data)

Searches for hidden patterns in data.

Input Set → Learning Algorithm → Clusters

Example

Images can be classified into:

- Vehicle → Bus, Truck, Scooter, etc.
- People
- Animal → Cat, Dog, etc.

In supervised learning:

- Continuous/Numerical output ⇒ **Regression**
- Categorical output ⇒ **Classification**

Examples

1. Given the features of a car, predict its price $\Rightarrow$ Regression.
2. Given an image of an animal, predict Cat or Dog $\Rightarrow$ Classification.
3. Given an image of a tumour, predict Cancerous or Not $\Rightarrow$ Classification.

Algorithms Used

Supervised Learning

- Linear Regression
- Logistic Regression
- KNN (K-Nearest Neighbour)
- Naïve Bayes
- Decision Tree (DT)
- etc.

Unsupervised Learning

- K-Means Clustering
- Hierarchical Clustering (Dendrogram)
- Expectation Maximization (EM) for Gaussian Mixture
- etc.

Data Type Example

Supervised Data

Features:

- Gender (X_1)
- Age (X_2)

Label:

- Health Status (Y): Sick / Healthy

Unsupervised Data

Contains only features (e.g., Gender and Age) without labels.